

WhatsApp Support Gateway - Video Recording Teleprompter

Silent-recording pacing guide with exact hold times per action for post-recording voiceover sync.

RECORDING STRATEGY (VIDEO FIRST, AUDIO LATER):

1. Start server: `uv run uvicorn gateway.main:app --host 0.0.0.0 --port 8000 --reload`
2. Open browser at `http://localhost:8000/demo` (100% zoom, both panes visible)
3. Follow each step's **ACTION**, then **STAY ON THAT SCREEN** for the indicated **HOLD TIME** before clicking the next chip!
4. Total video runtime: ~2 minutes 20 seconds (141 seconds).

STEP 0: INTRO & ARCHITECTURE [■■ HOLD ON SCREEN: 14s]

■ ACTION:

Hover mouse over WhatsApp phone on left, then live telemetry on right.

■ ON SCREEN:

Browser at `localhost:8000/demo` showing phone simulator & telemetry console.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 14s):

"Welcome to our WhatsApp Support Gateway demo. Built with Python and FastAPI, this system delivers sub-30ms webhook acknowledgment, secure database tool calling, and seamless live human takeover."

STEP 1: ORDER TRACKING (SUB-30MS & READ RECEIPTS) [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Click chip: 1. Track #ORD-1001

■ ON SCREEN:

3 green typing dots appear, double ticks turn blue (read receipt), latency <20ms, FedEx details returned.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"First, tracking an order: the webhook responds in under 20 milliseconds, marks the message as read with WhatsApp blue checkmarks, shows typing dots, and retrieves live FedEx status."

STEP 2: CONTEXT MEMORY & RETURN POLICY [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Click chip: 2. Can I return it?

■ ON SCREEN:

Bot resolves 'it' to #ORD-1001 and rejects return because shipment is in transit.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"Notice conversational context: asking 'Can I return it?' resolves 'it' to our active order. Because the package is still in transit, the bot strictly enforces return policy and explains returns require delivery first."

STEP 3: ANTI-IDOR SECURITY CHALLENGE [■■ HOLD ON SCREEN: 9s]

■ ACTION:

Click chip: 3. Track #ORD-1002 (Security Check)

■ ON SCREEN:

Bot intercepts query for #ORD-1002 (belongs to another phone), blocks details, and asks for last 4 digits.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 9s):

"For privacy, if a customer queries someone else's order, our Anti-IDOR layer intercepts, blocks details, and demands the last 4 digits on file."

STEP 4: OWNERSHIP AUTHENTICATION [■■ HOLD ON SCREEN: 7s]

■ ACTION:

Click chip: 4. Verify: 6543

■ ON SCREEN:

Bot confirms 'Security Verification Successful!' and unlocks #ORD-1002 status.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 7s):

"Entering the 4 digits validates ownership, unlocks the session, and presents the order status."

STEP 5: DELIVERED ORDER RETURN REQUEST [■■ HOLD ON SCREEN: 8s]

■ ACTION:

Click chip: 5. Return #ORD-1003

■ ON SCREEN:

Bot detects delivered order #ORD-1003 belongs to another account and requests 4-digit verification.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 8s):

"Now requesting a return on a delivered order: the gateway validates eligibility before authorizing return shipping."

STEP 6: PROGRAMMATIC PDF RETURN LABEL & BARCODE [■■ HOLD ON SCREEN: 15s]

■ ACTION:

Click chip: 6. Verify: 2233 (Get PDF)
Then click the PDF Document Card in phone.

■ ON SCREEN:

Bot approves return and delivers document card. Clicking opens vector PDF in new tab with Code128 barcode.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 15s):

"Once verified, the gateway programmatically generates a vector PDF return shipping label using ReportLab, complete with prepaid routing, a Code128 barcode, and an RMA packing slip."

STEP 7: SHOPIFY API INTEGRATION [■■ HOLD ON SCREEN: 11s]

■ ACTION:

(Switch back to demo tab) Click chip: 7. Shopify #1001

■ ON SCREEN:

Telemetry shows routing via ShopifyOrderAdapter. Bot returns live line items and fulfillment status.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 11s):

"Using clean ports and adapters, a single config flag switches from local storage to a live Shopify Admin REST API, seamlessly parsing live line items and fulfillment stages."

STEP 8: HUMAN ESCALATION & STRICT BOT MUTING [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Click chip: 8. Talk to Human

■ ON SCREEN:

Session State turns amber **ESCALATED_HUMAN**. External webhook fires. Operator Desk opens live WebSocket bridge.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"When a customer requests human help, an external webhook fires to Zendesk or n8n, strict bot muting engages, and our Operator Station opens a live WebSocket bridge."

STEP 9: BOT MUTING VERIFICATION (ZERO SPAM) [■■ HOLD ON SCREEN: 10s]

■ ACTION:

In phone chat input, type: Where is my refund? and send.

■ ON SCREEN:

Message appears on phone. Bot stays silent. Message appears live in Operator Chat Feed on right.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 10s):

"Customer follow-ups never trigger automated spam—the bot stays completely silent, while the customer's message streams live to our operator desk."

STEP 10: REAL-TIME 2-WAY OPERATOR CHAT [■■ HOLD ON SCREEN: 12s]

■ ACTION:

In Operator Desk, type: Hi, Agent Sarah here! Your refund of \$189.50 has been released. and click Send as Agent.

■ ON SCREEN:

Message appears immediately on customer phone with blue ■ Agent Sarah badge.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"As the human agent, typing a reply dispatches directly to the customer's WhatsApp in real time and logs to the permanent transcript."

STEP 11: SESSION RESOLUTION [■■ HOLD ON SCREEN: 7s]

■ ACTION:

Click button: Unmute Bot / Resolve

■ ON SCREEN:

Operator Station closes. Session State returns to green **ACTIVE_BOT**.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 7s):

"Once resolved, clicking 'Unmute Bot' re-enables automated bot handling for future inquiries."

STEP 12: PRODUCTION RIGOR & 94 TESTS [■■ HOLD ON SCREEN: 12s]

■ ACTION:

Switch to Terminal window and run: `uv run pytest`

■ ON SCREEN:

All **94 tests** pass **green** in under 1.5 seconds.

■■ SPOKEN SCRIPT (VOICEOVER FOR THESE 12s):

"Under the hood, the system is backed by Docker Compose, Redis Streams for horizontal scaling, strict mypy typing, and 94 automated tests passing in under 1.5 seconds."

Master Quick-Reference Cheat Sheet (With Wait Times)

Action (Click / Type)	■ ■ Hold Time	What Happens on Screen	Voiceover Script (To Read Aloud)
Hover mouse over WhatsApp phone on left, then live telemetry on right.	14s	Browser at localhost:8000/demo showing phone simulator & telemetry console.	"Welcome to our WhatsApp Support Gateway demo. Built with Python and FastAPI, this system delivers sub-30ms webhook acknowledgment, secure database tool calling, and seamless live human takeover."
Click chip: 1. Track #ORD-1001	12s	3 green typing dots appear, double ticks turn blue (read receipt), latency <20ms, FedEx details returned.	"First, tracking an order: the webhook responds in under 20 milliseconds, marks the message as read with WhatsApp blue checkmarks, shows typing dots, and retrieves live FedEx status."
Click chip: 2. Can I return it?	12s	Bot resolves 'it' to #ORD-1001 and rejects return because shipment is in transit.	"Notice conversational context: asking 'Can I return it?' resolves 'it' to our active order. Because the package is still in transit, the bot strictly enforces return policy and explains returns require delivery first."
Click chip: 3. Track #ORD-1002 (Security Check)	9s	Bot intercepts query for #ORD-1002 (belongs to another phone), blocks details, and asks for last 4 digits.	"For privacy, if a customer queries someone else's order, our Anti-IDOR layer intercepts, blocks details, and demands the last 4 digits on file."
Click chip: 4. Verify: 6543	7s	Bot confirms 'Security Verification Successful!' and unlocks #ORD-1002 status.	"Entering the 4 digits validates ownership, unlocks the session, and presents the order status."
Click chip: 5. Return #ORD-1003	8s	Bot detects delivered order #ORD-1003 belongs to another account and requests 4-digit verification.	"Now requesting a return on a delivered order: the gateway validates eligibility before authorizing return shipping."
Click chip: 6. Verify: 2233 (Get PDF) Then click the PDF Document Card in phone.	15s	Bot approves return and delivers document card. Clicking opens vector PDF in new tab with Code128 barcode.	"Once verified, the gateway programmatically generates a vector PDF return shipping label using ReportLab, complete with prepaid routing, a Code128 barcode, and an RMA packing slip."
(Switch back to demo tab) Click chip: 7. Shopify #1001	11s	Telemetry shows routing via ShopifyOrderAdapter. Bot returns live line items and fulfillment status.	"Using clean ports and adapters, a single config flag switches from local storage to a live Shopify Admin REST API, seamlessly parsing live line items and fulfillment stages."
Click chip: 8. Talk to Human	12s	Session State turns amber ESCALATED_HUMAN. External webhook fires. Operator Desk opens live WebSocket bridge.	"When a customer requests human help, an external webhook fires to Zendesk or n8n, strict bot muting engages, and our Operator Station opens a live WebSocket bridge."
In phone chat input, type: Where is my refund? and send.	10s	Message appears on phone. Bot stays silent. Message appears live in Operator Chat Feed on right.	"Customer follow-ups never trigger automated spam—the bot stays completely silent, while the customer's message streams live to our operator desk."
In Operator Desk, type: Hi, Agent Sarah here! Your refund of \$189.50 has been released. and click Send as Agent.	12s	Message appears immediately on customer phone with blue ■ Agent Sarah badge.	"As the human agent, typing a reply dispatches directly to the customer's WhatsApp in real time and logs to the permanent transcript."
Click button: Unmute Bot / Resolve	7s	Operator Station closes. Session State returns to green ACTIVE_BOT.	"Once resolved, clicking 'Unmute Bot' re-enables automated bot handling for future inquiries."
Switch to Terminal window and run: uv run pytest	12s	All 94 tests pass green in under 1.5 seconds.	"Under the hood, the system is backed by Docker Compose, Redis Streams for horizontal scaling, strict mypy typing, and 94 automated tests passing in under 1.5 seconds."